

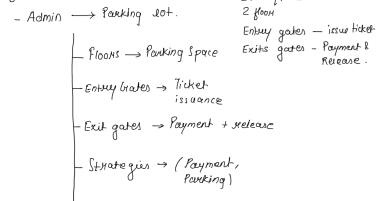
* Parking lot.

- vehicle
- Parking space
- Parking area for vehicle.
- Admin
- Ticket
- Payment
- Multiple floor structure.
- Multiple entry and exit gates.

Requirements :

- Vehicle type
- Dynamic slot Allocation
- Ticket created (At entry, ticket should be generated, ^{created,} At exit, user should be charged)
- Payment processing.
- Multiple floor support.
- Admin (Should be able to manage parking areas, gates and all).

High Level Architecture.



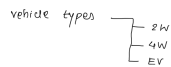
Control Plane → Admin configures floor/space/gates strategies.

Data Plane → Entry → Ticket → Exit.

Actors :

- Admin → configure infrastructure.
- vehicle owner → driving in / payment
- entry gates → Admit vehicles.
- exit gates → Payment / Release the space.

Q. Draw a UML Diagram showing the relationships and design patterns, principles and algos required.



Vehicle

- + Reg num
- + vehicle type

BikeVehicle, CarVehicle, Ev vehicle → subclasses.

